



The elephant in the room: Predictive performance of PLS models



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ABSTRACT

Attempts to introduce predictive performance metrics into partial least squares (PLS) path modeling have been slow and fall short of demonstrating impact on either practice or scientific development in PLS. This study contributes to PLS development by offering a comprehensive framework that identifies different dimensions of prediction and their effect on predictive performance evaluation with PLS. This framework contextualizes prior efforts in PLS and prediction and highlights potential opportunities and challenges. A second contribution to PLS development lies in proposed procedures to generate and evaluate different types of predictions from PLS models. These procedures account for the best practices that the new framework identifies. An outline of the many powerful ways in which predictive PLS methodologies can strengthen theory-building research constitutes a third contribution to PLS development. The framework, procedures, and research guidelines hopefully form the basis for a more informed and unified development of the rigorous theoretical and practical applications of PLS.

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1. Introduction and motivation

Business research has adopted PLS path modeling, known simply as PLSPM or just PLS, as a methodological tool that bridges exploration with explanation. The popularity of PLS stems partly from its ability to produce parameter estimates of complex models without many of the distributional and other constraints of traditional parametric methods. PLS achieves these estimates by using an iterative estimation procedure considered “predictive” in nature (Hair, Ringle, & Sarstedt, 2011). Yet, the PLS literature fails to fully exploit these predictive properties, focusing instead on explaining business phenomena from estimated model parameters that are assessed for statistical significance and power (Becker, Rai, & Rigdon, 2013). Researchers are now addressing this failure because they increasingly realize the shortcomings of assessing theoretical validity simply as statistical model fit—predictive performance must also be assessed (Armstrong, 2012; Woodside, 2013).

The importance of predictive performance extends beyond building purely predictive models to evaluating and improving the predictive power of explanatory models such as those built by PLS researchers (Shmueli, 2010). Evaluating predictive performance is recognized as useful for several theory building and validation purposes (Shmueli & Koppius, 2011): driving new theory; developing and evaluating measures and improving construct operationalization; comparing theories; improving existing theory; assessing relevance; and evaluating predictability of phenomena. This study addresses how to evaluate the

predictive power of explanatory PLS models for the purpose of improving theory. Evermann and Tate (2014) note that this approach is fundamental to the earliest visions of PLS. PLS path modeling was developed to occupy this middle ground and to straddle the traditional divide between causal-explanatory and predictive modeling at the extremes. It aims to maintain interpretability while engaging in predictive modeling. In fact, Herman Wold, who originally developed PLSPM, clearly and explicitly positioned it as a method for prediction (Dijkstra, 1983, 2010; Wold, 1982), de-emphasizing the importance of statistical tests and inference to population parameters.

Although many previous studies also proposed methods and metrics for evaluating predictive performance in PLS models, these efforts are largely disjoint and uninformed by the lessons learned in the field of predictive analytics. The most common predictive measure in the PLS literature is the in-sample squared correlation between estimated and true composite scores (e.g. Diamantopoulos, Sarstedt, Fuchs, Wilczynski, & Kaiser, 2012; Evermann & Tate, 2012; Henseler & Chin, 2010). A second metric is the bootstrapped version of the above statistic (Becker et al., 2013; Chin, 2010), wherein the estimated item weights of a fitted model are applied to bootstrapped samples to compute R^2 -based metrics. This second approach seems to recognize that a model might overfit a particular sample if predictive validity is evaluated on the same sample used for estimation; instead, predictions should use new data to avoid overfit. A third metric used as a measure of predictive validity is Q^2 (Chin, 2010; Evermann & Tate, 2012), which is based on blindfolding—a modified version of the jackknife approach. Lastly, Evermann and Tate (2014) suggest out-of-sample measures that are used in predictive analytics: Mean absolute error (MAE) and root mean squared error (RMSE), in which they aggregate case-wise prediction errors from a set of out-of-sample data.

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The multitude of prediction metrics and methods delivers a glimpse of the growing interest in demonstrating the predictive power of PLS models. But they also belie the lack of agreement on how to evaluate predictive performance in PLS and how this performance can inform theory building in practical ways. Fundamental to a proper predictive procedure is the ability to predict measurable information on new cases. Only this type of evaluation allows applying prediction to the future goals of empirical research. These goals include theory building and development based on the practical relevance of a model, assessing the predictability of phenomena, diagnosing measurement problems and potential improvements, identifying sources of unforeseen heterogeneity, evaluating model complexity, and reassessing theories in new contexts and over time. Only this type of evaluation supports the goal of deploying a PLS model for predicting measurable outcomes for new cases, as is often needed in practice.

This study is an attempt to cut through the confusion surrounding the notion of predictive performance in the context of PLS. This effort starts with the biggest elephant in the room: that predictive performance in PLS means being able to predict new data at the item-level. To appreciate this approach, three fundamental dimensions must be considered to properly understand the types of predictive performance and evaluations possible from PLS models and the benefits each of them confers. Only after this consideration can the achievements and shortcomings of prior efforts at prediction in the PLS literature be contextualized and the most promising directions for further developing tools, techniques, and metrics be identified. Such a vantage point permits the formation and proposal of a scheme to generate and evaluate predictions that apply the fundamental tenets of predictive analytics to the PLS context. This scheme can give rise to tools to which PLS practitioners and researchers can evaluate predictive performance to continuously re-assess and improve theory. The ultimate goal is to provide a framework and mindset that informs and unites future efforts to incorporate predictive and explanatory PLS methodologies.

2. A primer on prediction and PLS

2.1. Two key features of predictive performance in predictive analytics

As early as the 1950s, thinkers of the philosophy of science recognized the fundamental difference between explanation and prediction and the necessity of both (Dubin, 1969). Management research typically focuses on explanatory models. But with the recent popularity of Big Data and predictive analytics, some predictive modeling now appears in the management literature.

Before undertaking an examination of predictive performance in the context of PLS models, please note that in contemporary predictive analytics two key principles guide evaluation, as opposed to evaluating explanatory performance. First, at the data level, predictive modeling focuses on predicting new data, whereas explanatory modeling focuses on identifying and estimating a model that fits data at hand. This distinction further relates to the intended generalizations aimed by prediction versus explanation. The goal of predictive modeling is to predict the output value of new cases by applying the model parameters estimated from one data sample to generate predictions for individual cases outside of that sample. The primary goal of explanatory modeling is to test or quantify the underlying causal relationship between causes and effects that can be generalized from the sample to the population of interest.

Second, at the decision-making level, explanatory modeling focuses on macro decisions, whereas predictive modeling focuses on micro decisions. This difference requires explanatory modeling to look at an average case (e.g., the average household or the average customer), and predictive modeling looks at case-level predictions. Hence, an explanatory model with sufficient predictive power indicates its usability not only for making macro-level decisions but also for customized, micro-level decision making. For example, an explanatory model for customer

churn can explain the causes and outcomes of the average customer's attrition, but if the model also has sufficient predictive power, this model can be used for personalized predictions and recommendations. In contrast, another causal-explanatory model might indicate average employee attrition but have insufficient power to predict whether or not a particular individual will leave (Woodside, 2011).

2.2. Terminology and notation on PLS

Fig. 1 illustrates in schematic form the theoretical linkages in a PLS model. The paper will follow the terminology and notation of this simple model. The model has exogenous composites and an endogenous common factor, which establishes a unidirectional causal flow from input measures (x_{ij}) to outcome measures (y_{ij}). For full details on how to estimate a PLS model, please see Appendix A. The term “case” refers to a single observation or row of data. Discussions of individual cases use an index k added to the notation to represent the k -th case (x_{ijk} and y_{ijk}).

3. Predictive performance of PLS models: A conceptual framework

Unlocking the full potential of PLS predictions requires understanding the full picture of what is possible with predictions in PLS, where current proposals fit into this picture, and which of these proposals supports the goal of predictive performance evaluation of a PLS model.

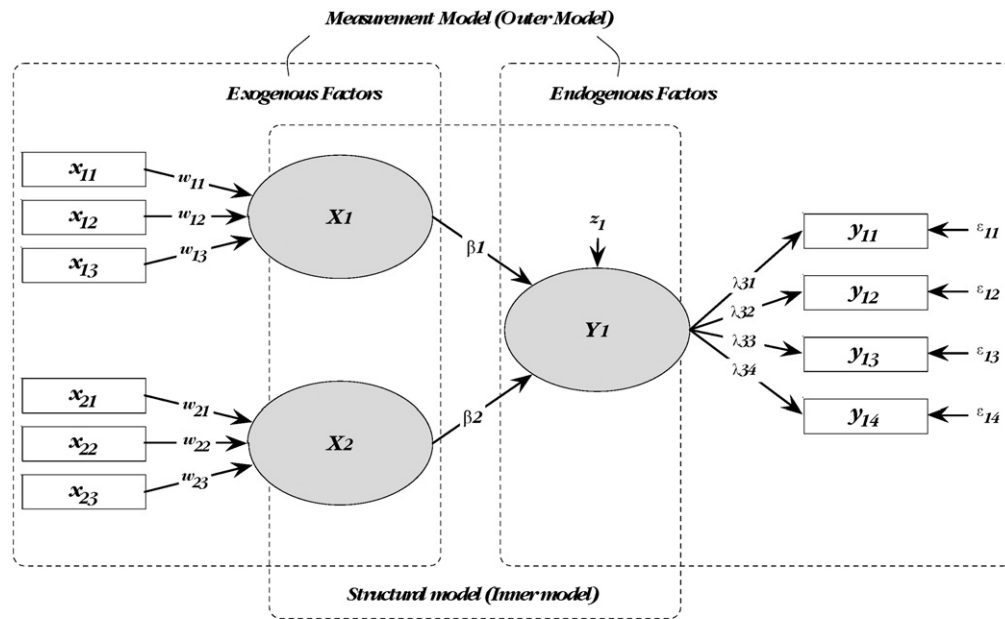
3.1. The three dimensions of predictive performance in PLS

Conceptualizing the possibilities of generating and evaluating predictions from PLS models requires considering three dimensions: predicting construct versus manifest items, predicting in-sample versus out-of-sample cases, and generating average-case predictions versus case-wise predictions. Fig. 2 depicts the three dimensions. The next section examines each of these dimensions.

3.1.1. Construct level versus item level (D1)

Understanding the various ways to use PLS models for prediction begins with contrasting construct-level and item-level prediction. This construct-versus-item dimension is a crucial first step to mapping the predictive landscape of PLS. Lohmöller's (1989) work serves as a guide to examining the various relational links between constructs and items in a PLS model. Like all efforts to explain causation, PLS models are essentially a statistical formalization of a theoretical causal linkage, derived from domain knowledge, between antecedent variables, and outcome variables (Kaplan, 2009). The theoretical linkage a PLS model introduces is itself a system of several smaller parametric models whose parameters define how manifest exogenous items relate to their exogenous factors (w), specify the structural paths between exogenous and endogenous factors (β), and relate endogenous factors to manifest outcome variables (λ). These various theoretical links within a PLS model each introduce a source of prediction error (d, z, ε) between the manifest items of exogenous factors (x_{ij}) and those of endogenous factors (y_{ij}). Thus, each link presents a smaller opportunity for prediction that can be used to gauge how that particular component of the theoretical model affects overall predictions. Fig. 3, adapted from Lohmöller (1989), illustrates five types of predictions a simple PLS model can generate.

The first three types of prediction that Lohmöller (1989) enumerates involve the fundamental measurement and structural characteristics of the theoretical model: predicting exogenous factor scores from their manifest items (*valid*), predicting endogenous factor scores from exogenous factor scores (*structural*), and predicting the items of exogenous factors from their factor scores (*communal*). By themselves, these three forms of prediction cannot give a full picture of the predictive performance of a model. For example, a model can have accurate *structural* predictions of endogenous factor scores (Y_i), and yet be unable to



Notes x_{ij} : Normalized indicators of exogenous factors, w_{ij} : Measurement weights, X_i : Scores of exogenous factors, β_i : Path coefficients, Y_i : Scores of endogenous factors, ζ_i : Error of endogenous factors, λ_{ij} : Factor loadings, y_{ij} : Normalized measures of endogenous factors

Fig. 1. A simple PLS model. Notes: x_{ij} : normalized indicators of exogenous factors, w_{ij} : measurement weights, X_i : scores of exogenous factors, β_i : path coefficients, Y_i : scores of endogenous factors, ζ_i : error of endogenous factors, λ_{ij} : factor loadings, y_{ij} : normalized measures of endogenous factors.

accurately predict outcome measures if the communality of the outcome items is poor. Likewise, a strong communality cannot lead to accurate predictions if structural predictive power is low.

More interesting prediction schemes arise when multiple parts of a model must interact to generate predictions. In particular, predicting the items of endogenous factors (y_{ij}) from the scores of exogenous factors (X_i) yields a sense of predictive *redundancy*. This procedure entails first predicting factor scores of endogenous variables (Y_i) as a prerequisite to predicting y_{ij} . In spirit, predictive redundancy allows seeing how the measured construct X_i predicts outcome factors and their corresponding measurements. Poor predictive redundancy should suggest that the outcome factors are ill-conceived when their structural constraints and item communalities are jointly considered. In principle, these results can be applied to reexamine the face validity of endogenous constructs and reexamine the theory that tied the exogenous and endogenous factors together.

Although not considered by Lohmöller, researchers have considered generating predictions of endogenous factor scores from the manifest items of exogenous factors (Becker et al., 2013). This type of prediction, which this study proposes to name *latent prediction*, is largely missing from practice in the applied PLS literature. Researchers have recently offered practical examples of such predictions. For example, PLS models of employee aggression could be used to rank-order or classify new employees by aggression factor scores (Chin, 2015).

Lastly, *operative prediction* entails using items of exogenous constructs (x_{ij}) to predict items of endogenous constructs (y_{ij}), and doing so by using the full measurement and structural properties posited by the PLS model. Operative prediction fully incorporates every aspect of the theoretical model in gauging predictive validity. Furthermore, operative prediction only requires manifest data items as predictors and outcomes. Only recently are researchers discussing operative prediction on

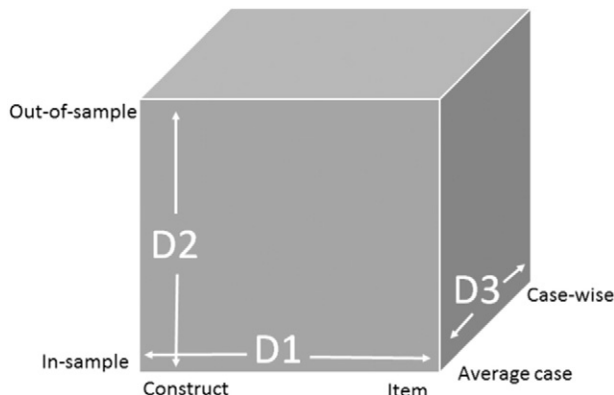


Fig. 2. The three dimensions of prediction in PLS.

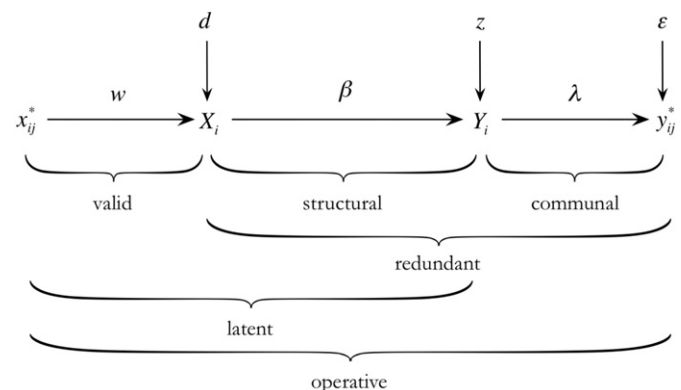


Fig. 3. Types of prediction from PLS models.

PLS models (Evermann & Tate, 2014; Shmueli & Koppius, 2010), although the literature lacks details or reviews of actual algorithms.

This distinction between items and constructs is unique to latent models such as PLS. The above typology linking all constructs and items helps determine how to evaluate predictive performance. If researchers are interested in evaluating PLS models in whole or in part, they must use either latent or operative prediction. These two forms of prediction from the above typology relate all factors and items to one another and thereby give a comprehensive sense of how each factor and item relates to predictions on the endogenous side of the model. Furthermore, these two types of prediction do not require prior knowledge of latent factor scores to generate predictions (see Fig. 4). This point becomes more salient with respect to performance evaluation when considering whether or not the predicted data is found within the sample used for estimation.

3.1.2. In-sample vs. out-of-sample (D2)

In-sample predictions (often called fitted values) can be useful for explanatory modeling efforts, in which the goal is to interpret the measurement and structural parameters of a PLS model. Generating in-sample predictions requires estimating the parameters of a PLS model by using a given data sample and then using the model to predict values for cases from the same sample (see Fig. 5). Note that the bootstrap, as is commonly used in PLS, is an in-sample approach because each bootstrap subsample is used both to estimate the model and to predict in-sample cases.

Out-of-sample predictions are useful for evaluation when the focus is on a model's ability to predict outcomes of heretofore unseen cases. Generating out-of-sample predictions involves estimating the model on a training sample that excludes the cases to be predicted. The model is then used to predict new cases (see Fig. 6). K-fold cross-validation is an out-of-sample approach: a sample is split into k parts, where $k-1$ parts are used as the training sample (for model estimation) and the k -th sample is designated out-of-sample. This process is repeated k times, each time designating a different subsample as the out-of-sample part.

Combining in-sample and out-of-sample predictions yields additional information about predictive performance. Whereas model fit and validity issues dominate quality concerns with explanatory modeling, the biggest danger to predictive performance is overfitting the model too tightly to the cases in the training data. An overfit model tends to perform poorly in predicting new cases. A comparison of the in-sample and out-of-sample prediction errors helps diagnose overfitting. A large discrepancy in favor of in-sample prediction is indicative of overfitting that likely translates to poor predictive performance in out-of-sample situations. The dichotomy between strong explanatory fit versus strong predictive power has two major ramifications for how predictive performance is evaluated. First, the explanatory estimation process and the predictive performance procedures must be done

separately because they involve different data and seek to minimize different criteria. Second, these two distinct evaluation processes must be judged by different metrics because high performance in one area does not necessarily bode well for the other.

Revisiting the distinction between item-level and construct-level predictions (D1) clarifies that only item-level data are amenable to out-of-sample predictive performance evaluation. Having both predictor and outcome variables as manifest items permits computing out-of-sample prediction errors. Estimated model parameters are used to compute factors scores, then predict outcome items and compare these predicted items to known actuals. In contrast, when either the predictors or the outcome are construct-level factor scores, prediction errors for out-of-sample cases are unobtainable, because the values for the actual factor scores are unknown. In the case of a construct-level outcome (e.g., *latent* prediction), although predicting endogenous factor scores for new cases is possible, actual factor score values are unknown, so prediction errors cannot be computed. In the case of construct-level predictors (e.g., *redundant* prediction), the outcome for new cases cannot be predicted because the input factor scores are unknown. Note that using predicted exogenous scores (using a *valid* type model) yields predicted scores that lack the variability associated with the noise term (d). Therefore, the resulting outcome predictions are less variable than they should be. In summary, if either the endogenous or exogenous variables are at the construct level, out-of-sample predictive performance cannot be evaluated, and in turn, the researcher cannot diagnose the generalizability of an existing model to new cases that were either not used or not available when the model was trained.

The above discussion reveals that the *structural*, *redundant*, and *communal* forms of prediction discussed earlier share a significant shortcoming. Because they require knowledge of the actual factor scores of each case in the data, these three types of prediction cannot use modern prediction schemes that require out-of-sample data cases for data partitioning or cross-validation, nor can they be applied to completely new cases.

Of the remaining two types of prediction identified in D1, only operative prediction can produce both in-sample and out-of-sample predictions and also evaluate predictions against known outcomes. By starting with manifest variables and generating predicted items, operative prediction schemes have all the qualities required for full out-of-sample evaluation. Note that valid and latent predictions are intermediate steps of the operative procedure. These two forms of predictions can be useful for in-sample and out-of-sample predictions because they start their prediction processes with manifest variables x_{ij} , but they cannot evaluate the power of their predictions because they produce only factor scores. To fulfill the goal of evaluating predictive performance, the framework presented in this paper rests on the premise that *predictive performance* is about the ability to predict wholly new data. Thus, the focus is on generating out-of-sample evaluations at the item level so that the training and holdout datasets are distinct, which is in line

		Predicted (Output)		
		X_i	Y_i	y_{ij}
Inputs	x_{ij}	Valid	Latent	Operative
	X_i		Structural	Redundant
	Y_i			Communal

Fig. 4. Types of predictions and the input information needed to generate them.

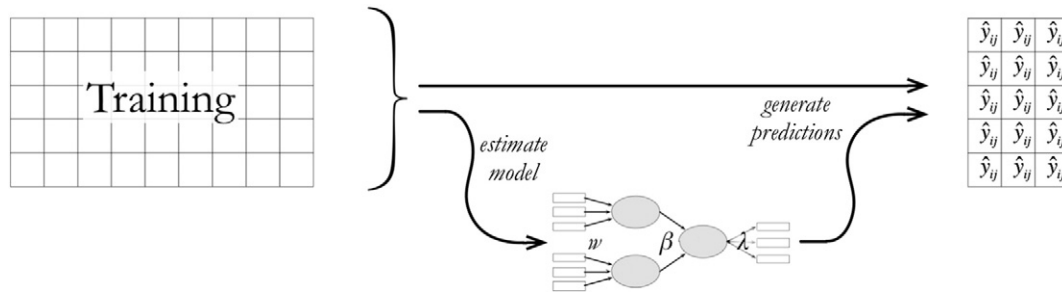


Fig. 5. In-sample prediction with PLS model.

with Armstrong (2012) and Woodside (2013). Metrics that clearly distinguish this notion of predictive performance from explanatory fit must also be sought.

3.1.3. Average case versus case-wise (D3)

The third dimension of prediction possibilities in PLS distinguishes between predicting the *average case* and each individual case (*case-wise prediction*). Predictive analytics typically focuses on case-wise prediction because the interest is in individual cases and in case-to-case heterogeneity. Predicting the average case is of interest when decisions are based on a prespecified grouping—for example, predicting outcomes for the average employed male under 30.

Regardless of whether predicting in-sample or out-of-sample, or looking at the item or the construct level, average-case prediction and case-wise prediction both yield the same point predictions. However, they differ in terms of the uncertainty associated with these point predictions. A case-wise prediction will be less precise (higher uncertainty) than an average-case prediction because of the need to also account for case-to-case variability. Therefore, average-case prediction intervals are narrower than case-wise prediction intervals. The terms *prediction accuracy* or *bias* ($\text{bias} = E(\text{prediction error})$) and *prediction precision* or *variance* ($\text{variance} = \text{Var}(\text{prediction error})$) help make this distinction clear: average case and case-wise predictions have the same accuracy level, but case-wise predictions will be less precise (see Appendix B).

The three dimensions of predictive performance in PLS, depicted as a cube in Fig. 2, lead to $2 \times 2 \times 2 = 8$ types of potential predictions. However, only two of the eight options allow for evaluating predictive performance in ways that support prediction-based theory building and development: item-level, out-of-sample, case-wise, or average-case prediction.

Nevertheless, construct-level, out-of-sample, case-wise prediction may also be of interest. An example of this type of prediction appears in Chin (2010): “If we develop a consumer based satisfaction scale to predict brand loyalty, will the weights derived to form the satisfaction scale be as predictive [for new cases]?” However, because out-of-sample factor score values are unavailable, evaluating the predictive performance of a PLS model in that sense would require a follow-up study in which actions based on the predicted out-of-sample case-

wise factor scores are compared with the outcomes from an alternative model or approach. For this reason, the predictive performance procedures present in Section 4 focus solely on item-level, out-of-sample prediction. Put simply, the focus is on predicting observable outcomes for new data. An important distinction between case-wise and average-case prediction is reflected in the predictive performance evaluation procedure and its implications.

3.2. The limitations and opportunities of existing approaches

The proposed conceptual framework and its three dimensions of how prediction can be performed using PLS models permits a reexamination of the common predictive performance methods and metrics used in the literature.

3.2.1. R^2 approach

Structural prediction errors are often diagnosed from the variance explained of endogenous variables as a multiple correlation coefficient (R^2). Even new prediction schemes proposed by Chin (2010) and Becker et al. (2013) favor prediction metrics based on R^2 . But although R^2 in PLS allows appreciation of the in-sample predictability of endogenous average-case factor scores, R^2 is not concerned with any of the other seven prediction options in Fig. 2, and specifically, does not capture the out-of-sample predictive performance, either case-wise or in average cases, of endogenous manifest items.

Recognizing the importance of out-of-sample evaluation, Becker et al. (2013) define and use an “out-of-sample R^2 ” metric that is based on case-wise, out-of-sample predictions of endogenous factor scores. Section 3.1 explains that because actual factor scores are unknown, out-of-sample latent predictions (factor scores) cannot be evaluated against actual values to produce prediction errors. Partly to overcome this limitation, Becker et al. (2013) follow a validation strategy proposed by Dana and Dawes (2004) wherein their training cases are a subset of their holdout cases. However, this approach faces the limitations discussed earlier with respect to the second dimension (D2) of PLS prediction: a holdout sample must be completely nonoverlapping with the training sample in order to minimize overfit and to allow assessment of true out-of-sample performance.

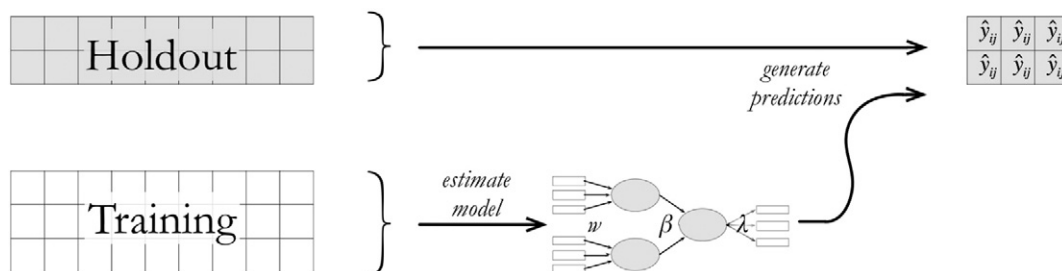


Fig. 6. Out-of-sample prediction with PLS model.

3.2.2. Q^2 approach

Another often used metric of predictive validity in PLS models is Q^2 , which is computed in two different ways to measure either *predictive communality* or, more commonly, *predictive redundancy* (Hair, Tomas, Hult, Ringle, & Sarstedt, 2014). Appendix C gives an overview of how Q^2 is computed. Q^2 gives a strong sense of redundancy within a dataset but nevertheless suffers many limitations from using highly correlated data from related items of the same construct and case as the items being predicted. First, the Q^2 procedure for PLS effectively conflates internal consistency with predictive accuracy because predictions of replaced items are heavily informed by the remaining items of the outcome construct. Second, the imputation method of Q^2 does not fully capture the type of heterogeneity associated with true out-of-sample prediction. This shortcoming occurs because by retaining all exogenous and most endogenous items of each case during training, Q^2 is not adequately blind to the variance associated with unusual cases in the dataset. Third, the Q^2 process only concerns itself with an aggregate sense of the predictability of a dataset instead of gauging the predictability of particular cases; software implementations of this procedure cannot produce case-wise predictions or prediction intervals for new cases. Finally, the Q^2 metric is itself difficult to evaluate in contextual terms. The guideline to interpret the positive values of Q^2 as indicating sufficient predictive relevance (Chin, 1998) is a rule-of-thumb that does not take into account the sampling distribution of Q^2 . Moreover, this scale-free statistic does not provide a sense of the magnitude of prediction errors. Thus, both Q^2 and its related “effect size” $q^2 = (Q_{included}^2 - Q_{excluded}^2)/Q_{included}^2$ are ad hoc metrics that do not provide highly interpretable results in terms of prediction error magnitude.

3.2.3. Operative prediction approach

Evermann and Tate (2014) propose a prediction scheme that overcomes many of the limitations of the approaches discussed above and that is in line with much of the discussion of the three dimensions of predictive evaluation in PLS. Their prediction scheme is case-wise, out-of-sample, and produces operative predictions at the item level. In practice, their approach should also be able to produce latent predictions for situations in which construct-level endogenous predictions are required, but predictive evaluation is not. Although Evermann and Tate's (2014) approach is in the right direction, several challenges must be overcome before truly useful and informative predictive evaluations can be obtained from PLS models. Although the next section deals with these shortcomings in detail, they can be summarized here.

To begin, the operative predictions from PLS are at the item-level, so the first challenge is whether to analyze each item separately or in combined form. Evermann and Tate (2014) generate a single RMSE metric from the multiple item-level RMSE values that operative prediction should yield. Importantly, any metric aggregated across items (on top of aggregation across cases) lacks sufficient detail to properly inform theory building. This uncertainty occurs because such metrics lack the granularity to identify both item-level and case-wise issues. Moreover, how can an evaluation based on a single item-aggregated metric scale to more realistic models with the added complexities of mediators or multiple endogenous factors? A related issue is the need for an approach to generate not only point predictions but also prediction intervals for both case-wise and average-case out-of-sample predictions.

Second, the way to perform normalization and denormalization with out-of-sample predictions from PLS models is not straightforward. Evermann and Tate (2014) test their approach against simulated normalized data but do not specify how practitioners should handle nonnormalized data that require care to avoid leaking descriptive training set information onto the holdout set. Third, studies proposing operative prediction schemes for PLS do not explain how practitioners can gauge the degree of overfitting of their models; such gauging requires comparing in-sample and out-of-sample predictive performance. Fourth, metrics for predictive evaluation have not been compared against naive benchmarks that do not consider any predictor

information. Fifth, no practical guidelines are available on how to use the results of predictive evaluation to improve theory. No formal specifications exist for how to generate and evaluate case-wise, out-of-sample, item-level predictions. Without such a specification, the PLS community cannot collectively replicate, review, and hone the relevant algorithms.

4. A procedure for evaluating predictive performance of a PLS model

The lack of established techniques and tools to generate and evaluate predictions from PLS has hindered progress on creating predictive methodology for PLS. This study proposes a set of procedures for prediction and evaluation that together create a framework for evaluating the predictive performance of PLS models. The framework is tailored to the desirable properties of predictive performance evaluation highlighted earlier: out-of-sample, item-level, and operative. Specifically, the procedure generates item-level case-wise and average-case predictions for both out-of-sample and in-sample cases. The out-of-sample behavior helps evaluate predictive performance on new data. And the comparison with the in-sample behavior serves as a diagnostic for possible overfitting of the PLS model to the training data.

Furthermore, this paper is accompanied by code for the free and open-source R statistical platform to implement these procedures. The code is available and maintained at <https://github.com/ISS-Analytics/pls-predict>. This software library will greatly aid PLS researchers who wish to incorporate predictive evaluation into their studies or to develop the predictive method further.

Table 1

Procedure for generating point predictions with a PLS model (item or construct level).

1. Split data into training and holdout sets of cases.
 2. Estimate parameters of PLS model with training data only: retain both initial descriptives of the training data (mean, standard deviation) and estimated measurement weights and loadings, and structural path coefficients.
 3. Initialize prediction variables from holdout set:
 - 3.1 Create the *Measurement Matrices X* and *Y* with holdout items of predictor and outcomes factors.
 - 3.2 Normalize measurements in *Measurements Matrices (X and Y)* using corresponding means and standard deviations from training data.
 - 3.3 Obtain *Inner Path Coefficients*, *Outer Weights*, and *Outer Loadings* from the estimated model and copy them into the *Inner Path Matrix (B)*, *Outer Weights matrix (W)*, and *Outer Loadings Matrix (L)* for the prediction process.
 - 3.4 Identify the endogenous outcome latent variables and their measurements to be predicted.
 4. Predict antecedent factor scores from Outer Weights: predict the factor scores of exogenous factors using normalized holdout measurements (*X*) and the *Outer Weights Matrix (W)* from the PLSPM model. Note that the *Measurements Matrix (X)* should have 0 values in positions in which factors and items do not correspond with one another. The new factor scores for antecedent variables are then calculated as $X_i = x_{ij} \times W_{ij}$.
 5. Loop until endogenous construct factor scores are estimated. With the antecedent factor scores, follow the path model forward, predicting the factors of each construct along the way, using the *path coefficients* from the training model.
For example, if X_1 and X_2 jointly lead to Y_1 , then the predicted factor scores of Y_1 are $Y_1 = X_1 \times \beta_1 + X_2 \times \beta_2$.
 6. Predict endogenous manifest items: once an endogenous construct, such as LV_3 , is reached, estimate the *predicted measurements matrix* (y_{ijk}), using the *outer loadings vector (L)*: $y_{ijk} = Y_{ik} \times L_{ij}$.
 7. Denormalize predictions: use the training data *mean* (μ) and *standard deviation* (σ) to bring the predictions back to the original scale. Multiply each predicted observation by its corresponding σ and add its corresponding μ .
$$y'_{ijk} = (y_{ijk} \cdot \sigma_{ij}) + \bar{E}_{ij}$$
- The resulting values are the case-wise and average-case point predictions (each case has as many predictions as the number of endogenous items).
8. Repeat steps 2–7 with the training data. The resulting values from step 5 are in-sample factor score predictions. The resulting values from step 7 are in-sample item-level predictions.

4.1. Proposed procedure for predicting with PLS models

4.1.1. Point predictions

The proposed procedure for generating point predictions (implemented in an algorithm called PLSpredict) is outlined in Table 1. The process can be extended to any PLS model, but the simple PLS model shown earlier illustrates the basics of the process (see Fig. 1 for terminology).

The procedure assumes a full training dataset of manifest variables (antecedent and outcome measurements) and a partial holdout (or prediction) dataset with new antecedent predictor measurements. Predictions for y_{ij} outcomes in the holdout (or prediction) dataset and also for the training set are generated (see Section 4.2 for their use in predictive performance evaluation). Note that predictions and prediction errors are generated for each item separately and are then used for item-level evaluation. During the development of the proposed prediction procedures, challenges arose regarding the measurement method of antecedent and outcome factors and the use of fully standardized parameter estimates to compute unstandardized outcomes. These challenges are described next.

4.1.2. Using manifest variables as predictors and outcomes

A key question that arises when considering prediction with a PLS model is which predictor and outcome variables to use when generating predictions from simple PLS models. The manifest variables of the most antecedent factors (e.g., X in the simple PLS model of Fig. 1) seem the most likely candidates for predictors, and the two candidates for outcomes are the measurement variables of endogenous factors and the latent scores of endogenous factors. However, the measurement model (common factor versus composite) of antecedent and outcome factors complicates these choices.

The predictors for the prediction process (Step 3) must capture both types of antecedents: composites and common factors. In the case of antecedent composites, their manifest items can be used as predictors. But in the case of antecedent common factors, factor scores are more appropriate predictors than manifest items, because of the latent nature of these factors. However, using the scores of common factors means that their manifest variables would simply be ignored in the prediction process and the resulting predictions would be invariant to the manifest items of common factors in the holdout set. Hence, the training weights of antecedent common factors are used to compute their scores as a weighted average, just as in the case of antecedent composites.

At first glance, using only these weights to determine the scores of exogenous common factors appears to contradict the latent nature of common factors. But recall that the iterative estimation process in the training stage computes weights at each step instead of loadings (Step 2.b.ii of the estimation process—see Appendix A) and that these weights are computed differently for composites versus common factors. This procedure implies that each factor's measurement mode is incorporated in its weights. Thus, the use of weights to compute factor scores for antecedent variables during the prediction process produces scores that are faithful to the specified measurement mode of each factor.

Similarly, when predicting measures of a purely endogenous outcome factor (Step 5), the predicted factor scores are multiplied by factor loadings, thereby effectively treating outcome measures as though they belonged to common factors. Note once again that differences in the estimation process for the two modes of measurement result in post hoc estimated loadings that differ for composites and common factors. Thus, using the estimated training loadings of all outcome factors to compute their measures should be a valid means of generating predictions that incorporate their measurement mode.

An alternative to predicting endogenous manifest items is to stop at predicting the factor scores of outcome composites. Although such predictions can have value (say, to rank-order cases by their predicted outcomes), the predictive validity of factor scores is difficult to ascertain because only the factor scores of the training set can be truly known.

Moreover, predicting manifest items helps address the challenge of linking measures to constructs (Woodside, 2011). Hence, the proposed prediction procedure uses antecedent manifest items to predict manifest items of outcome factors. The procedure does so by using weights to compute the scores of exogenous factors and loadings to compute the measures of endogenous outcome factors.

4.1.3. Descriptives for denormalization

The estimation process described earlier produces fully standardized, unit-free parameter estimates. Any prediction process that seeks to insert new predictors into an estimated PLS model must normalize these new predictors in advance (Step 2.2), and then denormalize the predicted outputs (Step 7). But what descriptive information to use to denormalize predicted outcome variables to their true scale is not necessarily clear. Available choices for denormalizing predictions are descriptive statistics (means and standard deviations) of the training data, holdout data, or both.

When the predicted values are part of a holdout dataset in which the actual outcomes are known, the most straightforward solution would be to denormalize the predicted outcomes by using the actual means and standard deviations of the observed outcome measures of holdouts. However, using descriptives from the holdout set might inadvertently strengthen the model's predictions by feeding in information from the holdout dataset. Furthermore, using holdout descriptives is clearly impossible for purely predictive situations in which outcomes have not yet been observed. Both issues were resolved in this study by denormalizing the predicted outcome manifest items using the mean and standard deviation of their training counterparts. This proposed solution is thus more faithful to the spirit of prediction because only the information discovered from training data is used (estimates of path coefficients, measurement weights and loadings, and statistical descriptives).

4.1.4. Prediction intervals

Although prediction efforts in PLS have concentrated thus far only on generating point predictions, generating prediction intervals (PIs) is also extremely useful. A PI specifies a range of values, with a given confidence level, in which the value is expected to fall. Prediction intervals convey the uncertainty associated with point predictions; wider intervals indicate higher uncertainty.

A PI for an average-case prediction accounts for the sampling variation associated with estimating the PLS parameters. A PI for a case-wise prediction takes into account two sources of variation: (1) the sampling variance associated with estimating the PLS parameters, and (2) the variation associated with the error term (see Appendix B). Such case-level information about uncertainty is useful for refining and developing more nuanced theoretical models. For example, finding that the uncertainty for some types of observations is higher than others might suggest a source of misfit that is not accounted for by theory.

Computing prediction intervals requires estimating the distribution of each prediction. Just as PLS models do not produce parameter statistics, this proposed point prediction process does not directly yield PIs. Instead, this study proposes a follow-up bootstrap approach, based on the *bootstrap percentile-t method* in Lam and Veall (2002) to obtain PIs for case-wise (or average-case) predictions. Specifically, to capture the sampling variation of the PLS coefficients requires generating a large number (B) of bootstrap samples from the training dataset (by sampling with replacement). A PLS model is estimated for each bootstrap sample $b = 1, \dots, B$, and that model is used to generate a point prediction for each of the holdout cases. The variation between the B point predictions for a holdout case captures the sampling variance associated with estimating the PLS parameters. This information is sufficient to produce a PI for an average-case prediction: a 90% (or in general, $(1 - \alpha) \cdot 100\%$) PI for a given point prediction can be computed as the $100 \cdot (\alpha/2) = 5$ th and $100 \cdot (1 - \alpha/2) = 95$ th percentiles of the B bootstrap point predictions.

Table 2

Steps for generating average-case and case-wise prediction intervals.

1. Create B bootstrapped samples from the training set (the size of each bootstrap sample is equal to the training set size n).
2. For each bootstrap sample b ($b = 1, \dots, B$):
 - 2.1 Estimate the PLS model from the bootstrap sample (step 2 of Table 1).
 - 2.2 Obtain point predictions for each item ij of case k in the holdout set y_{ijk}^b (steps 3–7 of Table 1).
 Note: We will end up with B point predictions for each y_{ijk}^b .
3. The $(1 - \alpha) \times 100\%$ PI for an average-case prediction is the interval between two relevant percentiles of the B point predictions y_{ijk}^b ($b = 1, \dots, B$).
4. For case-wise PI:
 - a. In each bootstrap sample b , compute the n prediction errors for item ij : $e_{ijk}^b = y_{ijk}^b - y_{ijk}$.
 - b. For each case k , item ij in bootstrap sample b , randomly draw a prediction error from $e_{ij1}^b, \dots, e_{ijn}^b$, and add the prediction error to y_{ijk}^b to obtain y_{ijk}^{*b} .
 - c. The $(1 - \alpha) \times 100\%$ case-wise PI is the interval between two relevant percentiles of the B values y_{ijk}^{*b} ($b = 1, \dots, B$).

Case-wise PIs require adding a bootstrap prediction error to each point prediction. For a point prediction generated in bootstrap sample b , randomly draw a prediction error from the corresponding bootstrap sample and add the error to the point prediction. The $(1 - \alpha) \times 100\%$ case-wise PI for {case i , item j } is then the $100 \times (\alpha/2)$ and $100 \times (1 - \alpha/2)$ percentiles of the B values of point prediction + error for that case. The detailed process appears in Table 2.

This process for producing prediction intervals is even more valuable for practitioners seeking to predict outcomes from new or simulated antecedent data because they would not have a holdout set or other basis for comparing predictions.

4.2. Evaluating predictive performance

Researchers seeking to understand the limits of theory should want to study the predictive performance of their explanatory models (Shmueli & Koppius, 2011). Three important principles of predictive evaluation include examining out-of-sample prediction errors and the metrics based on them, benchmarking the model predictions against a naive alternative, and assessing overfit. These evaluation issues are examined next.

4.2.1. Prediction metrics

As is typical in prediction studies, performance is analyzed based on prediction errors on a holdout sample and on summaries of these errors by using common prediction metrics such as root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). These metrics are suitable for numerical outcomes. However, each of these metrics has advantages and weaknesses in terms of what is captured and what is missed about error distribution. Consequently, researchers commonly examine distribution plots (histograms and boxplots) of the errors to see the full picture: the extent of over-versus under-prediction, outliers, variability of prediction errors, etc.

4.2.2. Evaluating predictive performance with small samples: Cross-validation

The classic data mining approach of evaluating predictive performance by extracting a holdout dataset may not be suitable for many applied PLS studies that have small- to medium-sized samples. Partitioning a small sample into training and holdout subsamples can introduce too much bias. In such cases, cross-validation is an alternative that allows assessment of predictive power by not reusing the same data for model estimation and prediction. In k -fold cross-validation, the data are randomly partitioned into k subsamples. The model is estimated using $k-1$ of the subsamples and predictions are generated for the remaining subsample which serves as a holdout set. This process is repeated k times to generate one prediction for each observation, from a model

Table 3

Generating and evaluating out-of-sample predictions using 10-fold cross-validation.

1. Split dataset and generate predictions for holdout subsamples:
 - 1.1. Split the dataset into 10 parts by randomly selecting rows (without replacement).
 - 1.2. For each part, fit the PLS model to the remaining 90% of data (serving training set) and predict the outcomes measurements of the remaining 10% (holdout set). Each of the n rows should have been a holdout at one point.
2. Recombine the results from the 10 folds and present predictive power:
 - 2.1. Combine predictions of all 10 parts (obtain predictions for all n records, separately for each item); compute prediction errors (residuals = actuals – predictions).
 - 2.2. Plot prediction vs. actual and prediction error plots and report summary statistics such as average error and RMSE.

estimated by using data that excluded the observation. Although any choice of k can be used, $k = 10$ is common in many predictive studies.

Thus, to evaluate the quality of predictions that can be generated by a PLS model, this study suggests a k -fold cross-validation process. The PLSpredict code developed as part of this research includes functions to perform k -fold cross-validation. Table 3 describes this procedure using the popular 10-fold validation. The evaluation algorithm produces errors for each prediction and overall measures of predictive accuracy (e.g., RMSE). This information, which will be discussed later, provides a perspective with which to diagnose theory from afar.

4.2.3. Naive benchmark

As is customary, evaluating the performance of a statistical model involves comparing a dependent variable as a function of input variables against a case in which no input information is available. In linear regression, for example, the R^2 measure compares model-based variability (SSE) with the overall variability of the dependent item (SST). Similarly, predictive modeling involves comparison of out-of-sample predictive performance with a naive benchmark that includes only information on the outcome (the endogenous items) but none of the predictor variables. The naive prediction for a particular manifest item of a PLS model is the simple item-level average in the training sample. The same prediction is used for every case. Using the metrics mentioned earlier (RMSE, MAPE, etc.) allows comparison of the performance of a predictive model to the naive benchmark.

4.2.4. Assessing overfitting

Although the separation of training and holdout samples is directed at avoiding overfitting, sometimes researchers try many different models (e.g., PLS models with and without a mediator, different numbers of items) and data configurations (e.g., removing outliers) when using the training sample. The result can be overfitting of the training data. A comparison of the out-of-sample prediction error distribution (and summary metrics) with in-sample counterparts can reveal if overfitting has occurred. A large discrepancy in favor of in-sample prediction is indicative of overfitting that will likely translate to poor predictive performance in out-of-sample situations.

In summary, the results of this study lead to proposed procedures to (1) generate case-wise and average-case out-of-sample item-level predictions from a PLS model with a training dataset and (2) evaluate out-of-sample predictive performance. Preliminary versions of these proposed procedures have been implemented in the R statistical environment. The full R code will be made available as an open-source repository under a GPL license. The efficacy of the proposed prediction procedures was tested by benchmarking predictions generated by PLSpredict for the simple model in Fig. 1 against a linear model that should generate similar predictions, given the straightforward nature of the simple model. Appendix D contains the tests and their results.

5. Enhancing PLS models using theory-informed prediction

As demonstrated, PLS models can be used not only to quantify and test a causal theory but also to generate point predictions and prediction intervals for manifest items both in-sample and out-of-sample. These predictions are based on a causal model, unlike predictions typically generated by machine learning algorithms. The ability of PLS to generate predictions and evaluate their performance gives PLS a self-diagnostic capability that permits use of predicted outcomes and metrics to evaluate and inform theory. Recall from the outset of this paper the six ways in which prediction can inform theory: driving new theory; developing and evaluating measures and improving construct operationalization; comparing theories; improving existing theory; assessing relevance; and evaluating predictability of phenomena (Shmueli & Koppius, 2011). The following sections discuss straightforward ways in which predictive performance evaluation of even simple PLS models can inform theory and propose a roadmap to build and compare theories in complex PLS models.

5.1. Using prediction to diagnose PLS models

Applying the procedures of prediction generation and evaluation from Section 4 to a simple PLS model, such as the one in Fig. 1, and a set of data can immediately yield basic feedback about the theory captured by the model.

The most immediate insight from predictive performance evaluation with PLS models relates to the predictability of phenomena. A model that predicts new cases no better than random prediction (e.g., its performance matches the naive benchmark) indicates that the phenomenon under investigation is unpredictable—because of a weakness in the theoretical model, due to the quality of the measured manifest items, or perhaps because the phenomenon is naturally unpredictable.

Another possible scenario from the predictive performance evaluation of a PLS model is that the model's ability to predict out-of-sample cases is better than random, but still insufficient for practical application. This situation can occur despite having explanatory strength. For example, survey items measuring employee or user behavior might have explanatory validity, but researchers might caution managers from directly applying such measures in practice if they have unacceptably low out-of-sample predictive accuracy or precision. A third scenario is a PLS model that has both explanatory validity and desirable out-of-sample prediction levels. If proposed outcomes can be strongly predicted, this predictability not only reinforces the theoretical value of a model but also suggests strong applicability. The last two scenarios highlight the uses of predictive performance for assessing the relevance

of theoretical PLS models as well as identifying opportunities to develop new theory.

The predictive evaluation of a PLS model may also be used to strengthen explanatory models by developing and evaluating measures to improve construct operationalization. In the presence of strong theory linking antecedent and outcome factors, the prediction errors of PLS models can help identify constructs that yield poor predictions and thus suggest that better measurement might improve face validity. Inadequate measurement of either antecedent or outcome phenomena may cause poor predictions, even when the factors themselves show good convergent and divergent validity. In this way, predictive performance can boost traditional rigorous measurement practices (e.g., Churchill, 1979) and become a key part of multi-item construct operationalization in PLS models.

The accuracy or precision of predictions can vary for different subgroups of observations. For some observations, the PLS predictions can be highly accurate, while for others, less accurate; for some subgroups, the model tends to over-predict while for others the model under-predicts. Detecting such “asymmetric relationships” (Woodside, 2013) or localized patterns can help in further theory development by creating more nuanced theories (e.g., by adding moderators or nonlinearities to the model) or identifying the boundaries of generalization around a theory. Furthermore, the prediction intervals proposed earlier can help improve existing theory. Case-level prediction intervals give a rare view of uncertainty at individual case and item levels rather than the broader average-case or construct-level understanding conveyed by parameter uncertainty. Special cases from prediction errors with extreme magnitude or from cases with unusually wide prediction intervals offer valuable insights. In such cases, researchers may consider interactions or missing variables that could help improve prediction accuracy and precision, thus improving the strength of theory and applicability alike.

Lastly, a scenario in which a PLS model shows weak explanatory validity but strong predictive performance presents an opportunity to develop new theory. A researcher might investigate why predictions are so accurate and whether their underlying correlations can be linked to a causal theory.

5.2. Using prediction to evaluate model complexity

Beyond diagnosing theoretical models, the predictive performance evaluation of PLS models can also be used to evaluate and compare models of growing levels of complexity. In particular, when mediating variables are inserted into PLS models, multiple prediction structures become possible. Consider, for example, the model in Fig. 7, which results from inserting a new factor (Y_1) into the earlier simple PLS model (see Fig. 1). This extended model is called the *complex model* to

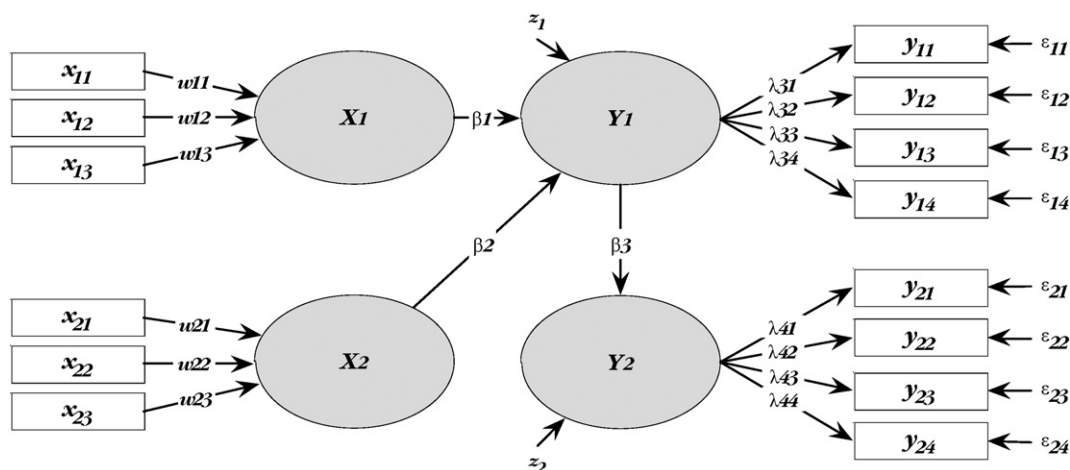


Fig. 7. Example of mediated PLS model.

be contrasted with the simpler PLS model considered thus far. The new factor (Y_1) in the complex model fully mediates the relationship between the antecedent factors (X_1 and X_2) and the outcome factor (Y_2). Note that the measurement weights and loadings of all factors in the complex model will differ from the estimates of the simple model because information about the new factor is now integrated into the measurements and paths of the model by the iterative estimation process of PLS. This new information introduced by the mediator plays an important role because now multiple schemes are possible for generating predictions with this complex model.

This study considers two straightforward prediction schemes for this complex model. The first prediction scheme uses the new mediator (Y_1) as the sole antecedent of the outcome (Y_2), so that the mediator's manifest items y_{1i} will be used to predict output items y_{2i} . Although this scheme seems to ignore the valuable role of X_1 and X_2 , recall that the training loadings of Y_1 include information about its antecedents. In this manner, Y_1 can be used as a parsimonious predictor of Y_2 .

An alternative prediction scheme is to follow the longer, indirect structural paths that originate from the measures of the earliest antecedents (x_{1i} and x_{2i}). Multiply these antecedents by their corresponding weights (w_{1i} and w_{2i}) to generate factor scores of X_i , then multiply these factor scores with their corresponding structural paths β_1 and β_2 to estimate factor scores for Y_1 , then multiply these factor scores by the structural estimate β_3 to get factor scores of Y_2 , and finally multiply Y_2 's factor scores with its loadings λ_{4i} to arrive at predictions for measures y_{2i} . This scheme fully utilizes the antecedent-most manifest items but now ignores measures y_{1i} of the mediator. Nonetheless, recall that the training weights, loadings, and structural paths relating to the antecedent and outcome factors incorporate information about the mediator.

These two prediction schemes are both straightforward, but each one avoids using some set of factors (either the antecedent-most factors or the mediator) in favor of another set of factors. But because both schemes contain some information about the factors they choose to omit, neither is completely blind to the predictor measures they omit. Researchers can use this duality to their advantage by generating predictions for outcome variables by using *both prediction schemes* and compare the quality of predictions to determine which factors are most meaningful in predicting outcomes. Full mediators (such as Y_1) should produce better predictions (lower out-of-sample RMSE or narrower prediction intervals) than antecedents. But if antecedents produce better-quality predictions, then researchers might want to reconsider the theoretical efficacy of their proposed mediators or acknowledge the shortcomings of their theory in terms of predictive performance. In this way, the alternative prediction schemes can be used to refine theorized models and augment traditional mediation tests such as Baron and Kenny's (1986) classical test of mediation and the Sobel test.

Finally, if predictions are to be used in practice and prediction accuracy is of importance, then a potentially useful approach is to combine predictions from different schemes (Armstrong, 2012). In the above example, predictions from the two schemes might be combined by taking a simple average of the two sets of predictions for new cases. Combined predictions (called *ensembles* in machine learning) are known to have higher predictive precision, especially when the predictions are negatively correlated, uncorrelated, or at least weakly positively correlated.

5.3. Reassessing a theory in new contexts and over time

Evaluation of the predictive performance of either simple or complex PLS models has even greater value when repeating such tests across settings or over time as new cases are added. For example, predictive evaluation of a PLS model can be used to determine whether a model is valid across settings such as geographic regions or industry

domains. Such an assessment certainly has value for researchers seeking to generalize theory but is also valuable for practitioners seeking to monitor the effectiveness of strategies they seek to implement in different settings. Predictive evaluation can also be repeated over time even within a given setting to diagnose whether predictions begin to improve or decline with maturation of a market or whether new trends are destabilizing previously established models.

The advantages of case-wise predictive performance measures over purely explanatory estimates and metrics (loadings, structural paths, variance explained, and bootstrapped statistics are again worth noting. Although models can be assessed across settings or time by using explanatory estimates, potential differences only indicate changes in the mean strength of relationships (i.e., how does the average-case change behavior over time). Predictive tests across settings and time that examine the prediction interval of each new case can also help determine the types of cases (or subjects) that display variant behavior across settings and time. Furthermore, new cases can be assessed as they are individually gathered, allowing for streaming real-time assessment of model validity. These benefits further enhance the usefulness of PLS models to practitioners who seek to roll out strategies captured by these models.

6. Conclusions and future directions

Being able to generate empirical case-wise out-of-sample predictions from a model and to evaluate the predictive power of explanatory models is vital to theory building and evaluation. Prediction using PLS models differs from machine learning algorithms and even from regression models in that PLS allows a researcher to incorporate a theoretically justified causal structure into the prediction process. Although regression models are useful for prediction, “[they are] most effective when dealing with small numbers of variables, and large amounts of reliable and valid data, where changes are expected to be large and predictable, and there are well-established causal relationships” (Armstrong, 2012). Nevertheless, Armstrong (2012) cautions against using regression models alone to specify causal relationships (the choice of variables to include, the expected direction of the relationship, etc.), and advocates an a priori approach that combines regression coefficients with values that reflect expert prior knowledge, similar to the Bayesian approach, but in a simpler fashion. PLS models provide such a framework that allows a researcher to combine a theory-driven model with data-based estimation and validation. PLS modeling appears to be more readily adopted than Bayesian modeling in social science research and practice because of the ease of representing a researcher's domain knowledge and theory in the model. For these reasons, expansion of PLS from mostly explanatory uses into the realm of explanation and predictive evaluation can lead to important contributions to research and to practice.

Generating and evaluating predictions of endogenous manifest items in PLS models can also help address criticism regarding concrete operationalization of constructs (Woodside, 2011). Although prediction by using PLS holds great promise, the process of generating empirical predictions with PLS poses both conceptual questions and technical challenges.

Armstrong (2012) emphasizes the need for software to “[make] it easy to simulate the forecasting situation.” This paper presents procedures (implemented in the algorithm PLSpredict) to generate and evaluate predictions with a PLS model from a set of data. This implementation is apparently among the first publicly available implementation that offers such level of detail. The proposed procedures and software implementation allow researchers not only to generate predictions but also to evaluate the predictive performance generated from their PLS models. The procedures (and PLSpredict algorithms) proposed here offer cross-validated case-wise and average-case point predictions, prediction errors, prediction error summaries (e.g., RMSE), and even prediction intervals. Implementation of these proposals is offered for

the open-source, free R statistical platform and easily accessible by researchers and practitioners. The procedures and frameworks presented in this paper can incorporate any PLS estimation algorithm, such as consistent PLS (PLSc) (Dijkstra & Henseler, 2015). The present study's proposals for procedures and tools form the foundation for the different ways in which predictive PLS can inform theory building in business research. Such a foundation breaks ground for an integrated explanatory-predictive methodological approach that jointly maximizes the explanatory power and applicability of business research.

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Appendix A. Review of PLS estimation

Although the algorithm for estimating PLS models is widely understood, this Appendix presents the algorithm using the same notation found throughout the main paper to help readers easily refer to particular steps of the process in the main paper.

PLS estimates both measurement weights and structural paths simultaneously as part of its iterative algorithm. The full method for estimating causal structures is discussed in greater detail in the literature (Barclay, Higgins, & Thompson, 1995; Lohmöller, 1989; Tenenhaus, Vinzi, Chatelin, & Lauro, 2005). Briefly, the estimation algorithm can be summarized as follows:

1. Initialization
 - a. Fully standardize all x_{ij} and y_{ij} across the n records of the given dataset.
 - b. Initialize all weights (w_{ij}) to 1.
 - c. Initialize X_i factor scores = $\Sigma(x_{ij})$; Y_i factor scores = $\Sigma(y_{ij})$.
2. Iteratively repeat:
 - a. Inner approximation:
 - i. Fully standardize X_i and Y_i scores by dividing by $\text{std}(X_i)$ and $\text{std}(Y_i)$, respectively.
 - ii. β_i coefficients (inner weights) are correlations between X_i and Y_i scores.
 - iii. Re-estimate X_i and Y_i scores based on estimated β_i .
 - b. Outer approximation:
 - i. Fully standardize X_i and Y_i scores.
 - ii. Estimate new outer weights using correlations for common factors, multiple regression for composites.
 - iii. Estimate X_i and Y_i scores by weighted x_{ij} and y_{ij} .
 - c. Check convergence (sum of weight-changes is smaller than a given threshold).
3. Estimate final results:
 - a. Estimate β_{XY} (weights of inner model) using multiple regression of Y_i over X_i s.
 - b. Estimate λ_{ij} (weights of outer model) using simple correlations.

This estimation algorithm is implemented in software such as the SmartPLS desktop application (Ringle, Wende, & Will, 2005) and the semPLS package (Monecke & Leisch, 2012) for the R statistical environment. The algorithm largely uses simple and multiple linear regressions

in an iterative manner until measurement weights and path coefficients converge.

Appendix B. Bias and variance of a PLS model

The terms *bias* and *variance* are used in machine learning with relation to a model. Consider an estimated model $E(Y) = f(x)$, where x can be a vector of input values. The expected prediction error (EPE) for a new case with value x_0 , using a quadratic loss function, is given by Hastie, Tibshirani, and Friedman (2009, page 223) as:

$$EPE = E\{Y - f(x_0)\}^2 = E\{Y - f(x_0)\}^2 + \{E(f(x_0)) - f(x_0)\}^2 + E\{f(x_0) - E(f(x_0))\}^2 = \text{Var}(Y) + \text{Bias}^2 + \text{Var}(f(x_0)).$$

The first term ($\text{Var}(Y)$) is the error that results even if the model is correctly specified and accurately estimated. *Bias* is the result of misspecifying the statistical model f . In the PLS setting, bias means specifying an incorrect structural model and/or measurement model. The third term, called *variance*, refers to sampling variance and is the result of using a sample to estimate f rather than having the exact population parameters. In PLS, the variance is the result of estimating the path parameters, weights, and loadings.

The above decomposition reveals a source of the difference between explanatory and predictive modeling: in explanatory modeling, the focus is on minimizing *bias* to obtain the most accurate representation of the underlying theory. In contrast, predictive modeling seeks to minimize the combination of *bias* and *variance*—that is why developing a pure prediction model occasionally intentionally introduces bias if the reduction in variance leads to an overall lower *bias*² + *variance*. In other words, pure predictive models might seek to sacrifice theoretical accuracy for improved empirical precision (Shmueli, 2010).

The decomposition also helps understand the difference in uncertainty levels between case-wise prediction and average-case prediction: whereas case-wise prediction is affected by all three terms in the above formula (the formula is in fact the variance of the case-wise prediction error), the average-case prediction is affected by only the last two terms.

Appendix C. The Q^2 procedure and metric

Q^2 is perhaps one of the more established prediction procedures in the PLS literature. Using details gathered from across several studies this Appendix provides a description of Q^2 in full algorithmic detail. This detailed description should provide a concise reference for PLS practitioners and researchers seeking to understand and apply Q^2 or seeking to compare and contrast Q^2 with other methods.

Q^2 for PLS is derived from a blindfolding procedure unique to PLS (Chin, 1998; Hair et al., 2014). This procedure is based on Stone–Geisser cross-validation methodology (Geisser, 1974; Stone, 1974) and is used to calculate the Q^2 metric proposed by Ball (1963). Briefly, the Q^2 procedure involves computing several blindfolded rounds on normalized data so that in each round one endogenous manifest item (y_{ij}^*) from each factor of each case is replaced by the average of the other manifest items of the same factor and case. More formally, if the omitted j -th item for case i is denoted by $y_{i(j)}^*$, then

1. Replace $y_{i(j)}^*$ with $\sum_j y_{ij}^* - y_{i(j)}^*$, so that a fraction $(1/d)$ of items have been replaced with even spacing between replaced items along columns and staggered across rows.
2. Estimate the PLS model with the imputed data.
3. Generate predictions for each (normalized) $y_{i(j)}^*$. These predictions, $y_{i(j)}^*$, are compared to the actual y_{ij}^* values to compute residuals: $y_{i(j)}^* - y_{i(j)}^*$.
4. Repeat earlier Steps (1–3) d times so that each item y_{ij}^* has been replaced exactly once.
5. Compare the sum of squared residuals $E = \sum_i \sum_j (y_{ij}^* - y_{i(j)}^*)^2$ with $O = \sum_i \sum_j (y_{ij}^* - \bar{y}_j^*)^2$, where $\bar{y}_j^*$ is the average of the j -th item, using the Q^2 statistic: $Q^2 = 1 - \frac{E}{O}$.

A key feature of the PLS Q^2 procedure is that in each round of computation, the algorithm replaces a single endogenous manifest item of each case with the mean of the other endogenous measurements for that case.

Appendix D. Evaluating and benchmarking PLSpredict

The quality of the PLSpredict procedure can be gauged by analyzing the its predictions. To this end, this appendix describes the application of PLSpredict to a simulated dataset, and a comparison of the resulting predictions to those from a linear model (LM). PLSpredict and LM are expected to produce very close predictions for simple models such as that in Fig. 1, where all antecedents directly affect an outcome.

D.1. Setting up benchmark and data methods

Various machine learning algorithms are designed to achieve high predictive power. Although any algorithm can be used as a predictive benchmark, benchmarking the predictive power of PLS against a method that allows the researcher to directly specify the input–output relationship is preferred. A good candidate is therefore the linear regression model (LM), which can be used for inference but is also a popular prediction tool. When using LM for prediction, the standard approach is to estimate regression parameters using training data and evaluate predictive performance by examining its predictions on a holdout sample. Prediction errors are the difference between the actual and predicted values.

In keeping with the tradition of prior methodological PLS studies (e.g., Sharma & Kim, 2013), this study uses a simulated dataset for benchmarking purposes to avoid any context or theory-specific vagaries that might favor one method over another. The *simsem* package (Pornprasertmanit, Schoemann, & Miller, 2014; Schoemann, Miller, Pornprasertmanit, & Wu, 2014) for the R statistical environment is a popular choice for simulating data for structural equation models as proposed by Vale and Maurelli (1983). Specifically, *simsem* is used to simulate data for the simple PLS example model shown in Fig. 1.

D.2. Predictions from PLS versus LM

The simulated dataset is used to verify that predictions based on PLS models do indeed generate meaningful predictions compared with other methods. Predictions from the PLS model are generated by using the PLSpredict algorithm proposed earlier and also with an LM model wherein all antecedent measures directly predict each outcome measure. In both cases, predictions are obtained for a holdout set. The point predictions of the four measures of the Y construct are shown in Fig. D1, where circles represent the PLS predictions and crosses represent LM predictions. The mean and standard deviations of the actual holdout outcome measures and those predicted by PLS and LM are also reported.

The predictions of the PLS and LM for a simple theoretic model are nearly identical in terms of their descriptives, mean, and standard deviation. Visually, the results show that the point predictions of PLS and LM are very similar, with the corresponding pairs of predictions from the two methods in close proximity to one another. Apart from descriptive and visual analysis, the prediction errors of both methods were also evaluated, as shown in Fig. D2. The standard deviation of prediction errors reported under each chart of Fig. D2 can be interpreted as RMSE. These values again suggest that great commonality exists between predictions generated by the simple PLS model and the LM predictions. Both methods appear to produce very similar errors of prediction for simple models.

These results empirically suggest that the proposed PLSpredict algorithm does indeed produce valid predictions. In particular, PLSpredict produces predictions very close to those from a more data-driven approach by using LM when the theoretical model in question is a simple one with very minimal theoretical constraints. However, the predictions of LM and PLSpredict will not always be the same. For example, if antecedent and outcome factors of the PLS model remain the same but the structural paths change, or if a mediating structure is introduced, the LM predictions will remain the same but predictions from PLSpredict will likely diverge. Because the predictions of PLSpredict are sensitive to the theory of the PLS model, PLSpredict can be used to diagnose and evaluate theory in multiple ways.

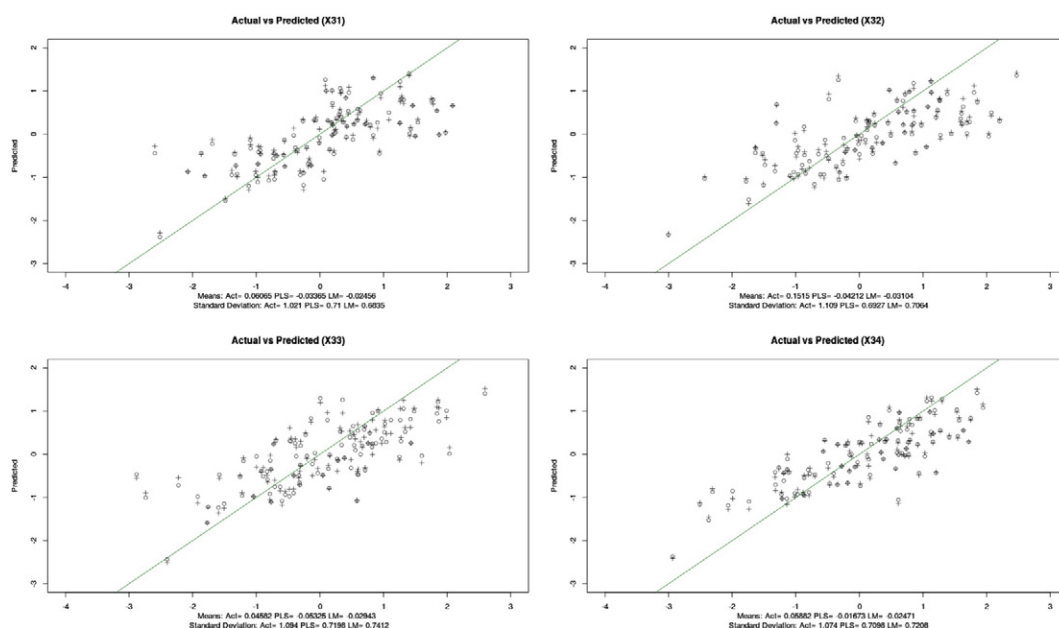


Fig. D1. Outcomes—holdout actuals versus PLS predictions versus LM predictions. Notes: Each of the four charts show: point predictions generated by PLS (circle) and LM (cross); diagonal line represents prediction = actual; labels showing descriptives (mean and standard deviation) of actual holdout outcomes, PLS predictions, and LM predictions.

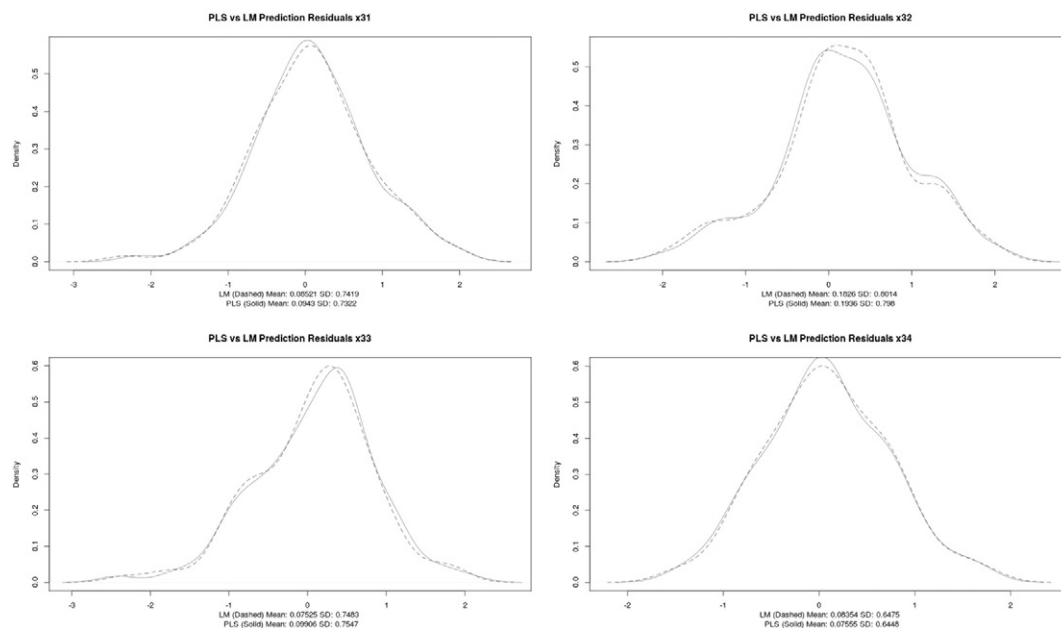


Fig. D2. Prediction errors of PLS versus LM. Notes: Each of the four charts show: density curve of prediction errors (actual–prediction) from PLS (solid) versus LM (dashed); labels below each chart report descriptives (mean and standard deviation) of PLS versus LM prediction errors. Standard deviation of prediction errors can be interpreted as RMSE.

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